

# Topological insulator --- Solution

Y26 (2026) — English translation

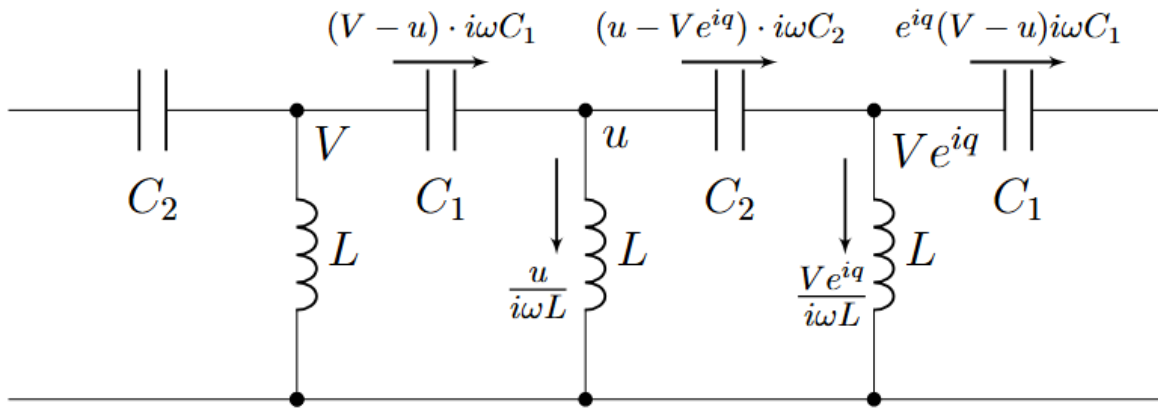
**A1** Using Kirchhoff's rules, obtain a system of equations relating  $V$ ,  $u$  and  $q$ . Express your answer using  $V$ ,  $u$ ,  $q$ ,  $L$ ,  $C_1$ ,  $C_2$  and  $\omega$ . **0.40**

Let us find the currents flowing in the links, knowing that at the point located to the right of the point  $C$  the potential is equal to  $ue^{iq}$ .

Now let's write the first Kirchhoff rule for nodes with potentials  $u$  and  $Ve^{iq}$ , respectively

$$\begin{cases} (V - u) \cdot i\omega C_1 = (u - Ve^{iq}) \cdot i\omega C_2 + \frac{u}{i\omega L}, \\ (u - Ve^{iq}) \cdot i\omega C_2 = e^{iq}(V - u) \cdot i\omega C_1 + \frac{Ve^{iq}}{i\omega L}. \end{cases}$$

The resulting system of equations is sufficient to obtain an expression for  $\cos q(\omega)$ .



**A2** From the resulting system, express  $\cos q$  in terms of  $\omega$ ,  $L$ ,  $C_1$  and  $C_2$ . **0.80**

Method 1. From both equations we express  $\frac{u}{V}$  and equate them. We obtain from equations 1 and 2, respectively:

$$\begin{cases} \frac{u}{V} = \frac{e^{iq} \cdot i\omega C_2 + i\omega C_1}{i\omega \cdot \left( C_1 + C_2 - \frac{1}{\omega^2 L} \right)}, \\ \frac{u}{V} = \frac{e^{iq} \cdot i\omega \cdot \left( C_1 + C_2 - \frac{1}{\omega^2 L} \right)}{e^{iq} \cdot i\omega C_1 + i\omega C_2}, \end{cases}$$

from where, equating  $\frac{u}{V}$  from different equations to each other and reducing by  $i\omega$ , we obtain an equation containing only the unknown  $q$ :

$$(e^{iq}C_2 + C_1)(e^{iq}C_1 + C_2) = e^{iq}\left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2.$$

Теперь раскроем скобки слева и поделим обе части уравнения на  $e^{iq}$  :

$$C_1^2 + C_2^2 + C_1C_2(e^{iq} + e^{-iq}) = \left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2.$$

Учитывая, что  $e^{iq} + e^{-iq} = 2\cos(q)$ , получаем соотношение на  $\cos q$

$$\cos q = \frac{\left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2 - C_1^2 - C_2^2}{2 \cdot C_1C_2}.$$

Метод 2. На самом деле, быстрее поступить немного иначе. Запишем систему уравнений как линейную относительно  $V$  и  $u$ ,

$$\begin{cases} V(C_1 + C_2e^{iq}) + u\left(-C_1 - C_2 + \frac{1}{\omega^2 L}\right) = 0, \\ V\left(-C_1 - C_2 + \frac{1}{\omega^2 L}\right) + u(C_1 + C_2e^{-iq}) = 0. \end{cases}$$

Чтобы система имела нетривиальное решение (их бесконечное количество), необходимо и достаточно приравнять нулю главный определитель системы. Таким образом,

$$(C_1 + C_2e^{iq})(C_1 + C_2e^{-iq}) = \left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2.$$

Окончательно,

$$\cos q = \frac{\left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2 - C_1^2 - C_2^2}{2 \cdot C_1C_2}.$$

We obtain from equations 1 and 2, respectively:

$$\begin{cases} \frac{u}{V} = \frac{e^{iq} \cdot i\omega C_2 + i\omega C_1}{i\omega \cdot \left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)}, \\ \frac{u}{V} = \frac{e^{iq} \cdot i\omega \cdot \left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)}{e^{iq} \cdot i\omega C_1 + i\omega C_2}, \end{cases}$$

whence, equating  $\frac{u}{V}$  from different equations to each other and canceling by  $i\omega$ , we obtain an equation containing only the unknown  $q$ :

$$(e^{iq}C_2 + C_1)(e^{iq}C_1 + C_2) = e^{iq}\left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2.$$

Now let's open the parentheses on the left and divide both sides of the equation by  $e^{iq}$  :

$$C_1^2 + C_2^2 + C_1C_2(e^{iq} + e^{-iq}) = \left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2.$$

Considering that  $e^{iq} + e^{-iq} = 2 \cos(q)$ , we obtain the relation for  $\cos q$

$$\cos q = \frac{\left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2 - C_1^2 - C_2^2}{2 \cdot C_1 C_2}.$$

Method 2. In fact, it's faster to do things a little differently. Let us write the system of equations as linear with respect to  $V$  and  $u$ ,

$$\begin{cases} V(C_1 + C_2 e^{iq}) + u\left(-C_1 - C_2 + \frac{1}{\omega^2 L}\right) = 0, \\ V\left(-C_1 - C_2 + \frac{1}{\omega^2 L}\right) + u(C_1 + C_2 e^{-iq}) = 0. \end{cases}$$

For a system to have a nontrivial solution (there are an infinite number of them), it is necessary and sufficient to set the main determinant of the system equal to zero. Thus,

$$(C_1 + C_2 e^{iq})(C_1 + C_2 e^{-iq}) = \left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2.$$

Finally,

$$\cos q = \frac{\left(C_1 + C_2 - \frac{1}{\omega^2 L}\right)^2 - C_1^2 - C_2^2}{2 \cdot C_1 C_2}.$$

**A3 Determine at what frequencies the wave vector  $q$  will have an imaginary part. Express your answer in terms of  $L$ ,  $C_1$  and  $C_2$ . 0.60**

First way. Let us give an expression for  $\cos q$

$$\cos q = 1 - \frac{C_1 + C_2}{C_1 C_2} \cdot \frac{1}{\omega^2 L} + \frac{1}{2C_1 C_2} \cdot \left(\frac{1}{\omega^2 L}\right)^2$$

Решим два квадратных неравенства

$$\begin{aligned} \cos q &> 1; \\ -\frac{C_1 + C_2}{C_1 C_2} \cdot \frac{1}{\omega^2 L} + \frac{1}{2C_1 C_2} \cdot \left(\frac{1}{\omega^2 L}\right)^2 &> 0; \\ \frac{1}{\omega^2 L} \left(\frac{1}{\omega^2 L} - 2(C_1 + C_2)\right) &> 0; \\ \frac{1}{\omega^2 L} &\in (-\infty; 0) \cup (2(C_1 + C_2); +\infty); \\ \omega^2 L &\in \left(0; \frac{1}{2(C_1 + C_2)}\right) \end{aligned}$$

$$\begin{aligned}
& \cos q < -1; \\
& 2 - \frac{C_1 + C_2}{C_1 C_2} \cdot \frac{1}{\omega^2 L} + \frac{1}{2C_1 C_2} \cdot \left( \frac{1}{\omega^2 L} \right)^2 < 0; \\
& \left( \frac{1}{\omega^2 L} \right)^2 - 2(C_1 + C_2) \cdot \frac{1}{\omega^2 L} + 4C_1 C_2 < 0; \\
& \left( \frac{1}{\omega^2 L} - 2C_1 \right) \left( \frac{1}{\omega^2 L} - 2C_2 \right) < 0; \\
& \frac{1}{\omega^2 L} \in (2C_1; 2C_2); \\
& \omega^2 L \in \left( \frac{1}{2C_1}; \frac{1}{2C_2} \right)
\end{aligned}$$

Из объединения системы неравенств получаем конечный ответ

$$\begin{aligned}
& \omega \in \left[ 0; \frac{1}{\sqrt{2(C_1 + C_2)L}} \right) \cup \left( \frac{1}{\sqrt{2C_2L}}; \frac{1}{\sqrt{2C_1L}} \right). \\
& \cos q = 1 - \frac{C_1 + C_2}{C_1 C_2} \cdot \frac{1}{\omega^2 L} + \frac{1}{2C_1 C_2} \cdot \left( \frac{1}{\omega^2 L} \right)^2
\end{aligned}$$

Let's solve two quadratic inequalities

From combining the system of inequalities we obtain the final answer

$$\omega \in \left[ 0; \frac{1}{\sqrt{2(C_1 + C_2)L}} \right) \cup \left( \frac{1}{\sqrt{2C_2L}}; \frac{1}{\sqrt{2C_1L}} \right).$$

Second way. Let's write down the expression for  $x = \cos q$

$$x = 1 - \frac{C_1 + C_2}{C_1 C_2} \cdot \frac{1}{\omega^2 L} + \frac{1}{2C_1 C_2} \cdot \frac{1}{\omega^4 L^2}.$$

Получаем, что зависимость  $x\left(\frac{1}{\omega^2 L}\right)$  квадратичная, построим ее график качественный график, на котором отметим точки пересечения с прямыми  $\cos q = 1$  и  $\cos q = -1$ .

The resulting graph can be qualitatively reconstructed in the  $y = \cos q$   $x = \omega^2 L$  axes, preserving all extrema and transforming the  $x$  axis.

Let us introduce the notation corresponding to the coordinates  $x$  at the point of intersection of the constructed graph with the lines  $\cos q = 1$ ,  $\cos q = -1$ . At the point of intersection of the graph with  $\cos q = 1$ ,  $\omega = \omega_1$ . At the intersection point of the graph with  $\cos q = -1$ ,  $\omega = \omega_2$  and  $\omega = \omega_3$ . Let's compose and solve quadratic equations for  $\frac{1}{\omega^2 L}$

$$\begin{aligned}
& \cos q = 1; \\
& -\frac{C_1 + C_2}{C_1 C_2} \cdot \frac{1}{\omega^2 L} + \frac{1}{2C_1 C_2} \cdot \left( \frac{1}{\omega^2 L} \right)^2 = 0; \\
& \frac{1}{\omega^2 L} \left( \frac{1}{\omega^2 L} - 2(C_1 + C_2) \right) = 0; \\
& \frac{1}{\omega^2 L} = 2(C_1 + C_2) \text{ or } \frac{1}{\omega^2 L} = 0
\end{aligned}$$

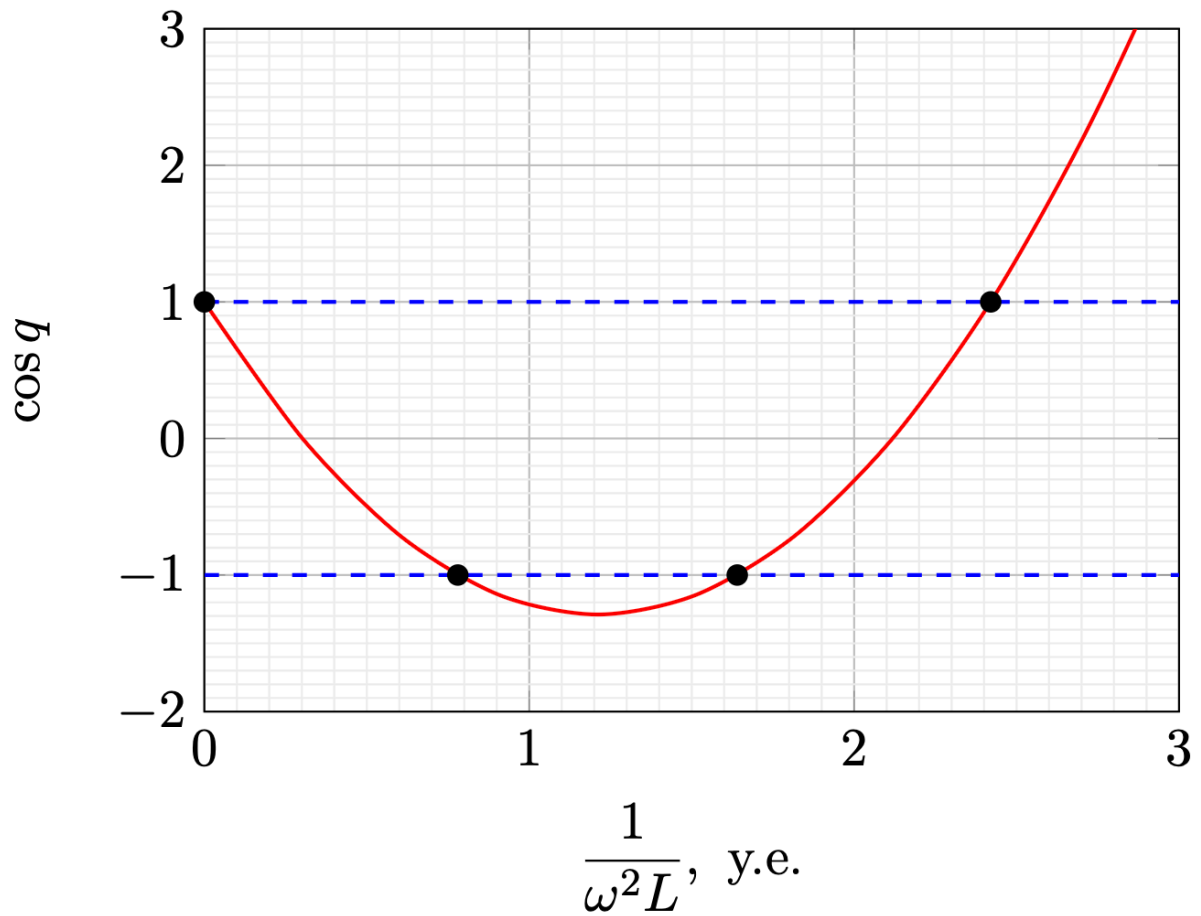
$$\begin{aligned}
& \cos q = -1; \\
& 2 - \frac{C_1 + C_2}{C_1 C_2} \cdot \frac{1}{\omega^2 L} + \frac{1}{2C_1 C_2} \cdot \left( \frac{1}{\omega^2 L} \right)^2 = 0; \\
& \left( \frac{1}{\omega^2 L} \right)^2 - 2(C_1 + C_2) \cdot \frac{1}{\omega^2 L} + 4C_1 C_2 = 0; \\
& \left( \frac{1}{\omega^2 L} - 2C_1 \right) \left( \frac{1}{\omega^2 L} - 2C_2 \right) = 0; \\
& \frac{1}{\omega^2 L} = 2C_1 \text{ or } \frac{1}{\omega^2 L} = 2C_2
\end{aligned}$$

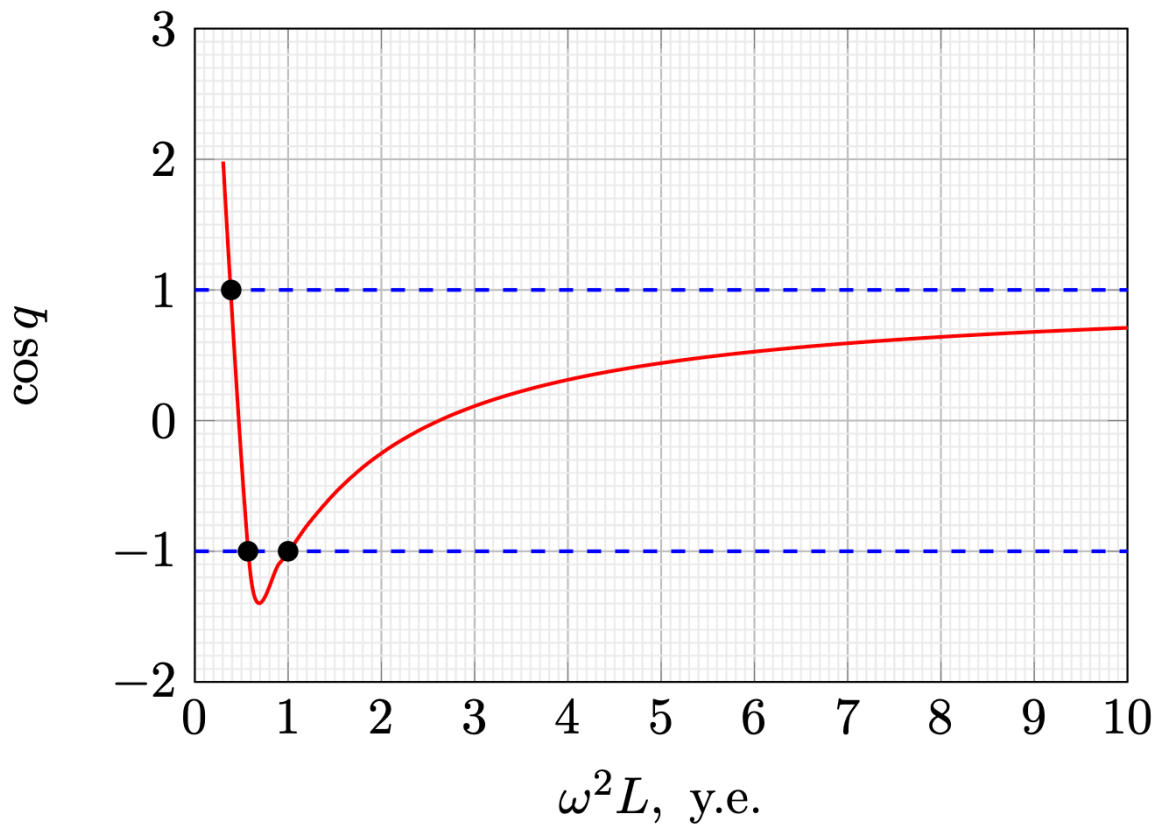
Comparing the solutions of the equations with the graph, we get the answer

$$\omega \in \left[ 0; \frac{1}{\sqrt{2(C_1 + C_2)L}} \right) \cup \left( \frac{1}{\sqrt{2C_2L}}; \frac{1}{\sqrt{2C_1L}} \right).$$

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$$\omega \in \left[ 0; \frac{1}{\sqrt{2(C_1 + C_2)L}} \right) \cup \left( \frac{1}{\sqrt{2C_2L}}; \frac{1}{\sqrt{2C_1L}} \right).$$





**A4** Measure the dependence of the real and imaginary components of the wave vector  $\Re(q)$  and  $\Im(q)$  on the frequency  $f$ . Describe the measurement technique. Take measurements in the frequency range from 650 кГц to 1300 кГц in steps of 50 кГц. For each frequency  $f$ , the value of both  $\Re(q)$  and  $\Im(q)$  must be obtained. **1.50**

Let's connect the generator to the circuit according to the instructions in the condition. Using two oscilloscope channels, we will measure the doubled voltage amplitude on the generator  $U_0$  and the doubled voltage amplitude on the first link  $U_1$ . Using the phase shift of the signals on an oscilloscope we will measure  $\Re(q)$

$$\Re(q) = 2\pi f \Delta T$$

По отношению амплитуд на сигналах будем измерять  $\Im(q)$

$$\Im(q) = \ln \frac{U_0}{U_1}$$

Примечание. При проведении измерений в диапазоне частот  $f \in [770; 890]$  kHz цепь находится в проводящем состоянии (соответствует диапазону частот  $f \in \left( \frac{1}{\sqrt{2(C_1 + C_2)L}}; \frac{1}{\sqrt{2C_2L}} \right)$ ), therefore, during measurements, the influence of the reflected wave can be noticed. When linearized, these points will produce outliers.

Using the phase shift of the signals on an oscilloscope we will measure  $\Re(q)$

$$\Re(q) = 2\pi f \Delta T$$

Based on the ratio of the amplitudes on the signals, we will measure  $\Im(q)$

$$\Im(q) = \ln \frac{U_0}{U_1}$$

Note. When making measurements in the frequency range  $f \in [770; 890]$  кГц, the circuit is in a conducting state (corresponding to the frequency range  $f \in \left( \frac{1}{\sqrt{2(C_1 + C_2)L}}; \frac{1}{\sqrt{2C_2L}} \right)$ ), so during measurements you can notice the influence of the reflected wave. When linearized, these points will produce outliers.

$f$ , кГц  $\Delta T$ , нс  $U_0$ , мВ  $U_1$ , мВ  $\Re(q)$   $\Im(q)$   $f^{-2}$ ,  $10^{-12}$  с<sup>2</sup>  $\Re(\cos q)$   $(1 - \cos q) \cdot f^2$ ,  $10^{12}$  Гц<sup>2</sup>

|      |       |       |      |       |       |       |       |       |      |       |       |      |       |       |       |       |       |      |       |       |     |      |      |
|------|-------|-------|------|-------|-------|-------|-------|-------|------|-------|-------|------|-------|-------|-------|-------|-------|------|-------|-------|-----|------|------|
| 650  | 0     | 57.6  | 8    | 0.00  | -1.97 | 2.37  | 3.67  | -1.13 | 700  | 32    | 128   | 34.4 | 0.14  | -1.31 | 2.04  | 1.98  | -0.48 | 750  | 216   | 284   | 408 | 1.02 | 0.36 |
| 1.78 | 0.56  | 0.25  | 800  | 452   | 616   | 416   | 2.27  | -0.39 | 1.56 | -0.70 | 1.09  | 850  | 396   | 328   | 172   | 2.11  | -0.65 | 1.38 | -0.63 | 1.18  |     |      |      |
| 900  | 480   | 304   | 212  | 2.71  | -0.36 | 1.23  | -0.97 | 1.60  | 950  | 516   | 252   | 134  | 3.08  | -0.63 | 1.11  | -1.21 | 1.99  | 1000 | 496   | 232   |     |      |      |
| 112  | 3.12  | -0.73 | 1.00 | -1.28 | 2.28  | 1050  | 484   | 232   | 110  | 3.19  | -0.75 | 0.91 | -1.29 | 2.53  | 1100  | 460   | 244   | 120  | 3.18  | -0.71 |     |      |      |
| 0.83 | -1.26 | 2.74  | 1150 | 436   | 268   | 142   | 3.15  | -0.64 | 0.76 | -1.21 | 2.92  | 1200 | 416   | 304   | 182   | 3.14  | -0.51 | 0.69 | -1.13 |       |     |      |      |
| 3.07 | 1250  | 396   | 368  | 264   | 3.11  | -0.33 | 0.64  | -1.06 | 3.21 | 1300  | 388   | 520  | 464   | 3.17  | -0.11 | 0.59  | -1.01 | 3.39 |       |       |     |      |      |

**A5** Linearize the dependence of  $\Re(\cos q)$  on the frequency of the generator  $f$  and determine the coefficients of the linear dependence. **1.00**

Using the result of the previous points

$$\cos q = 1 - \frac{C_1 + C_2}{C_1 C_2} \cdot \frac{1}{\omega^2 L} + \frac{1}{2C_1 C_2} \cdot \frac{1}{\omega^4 L^2}.$$

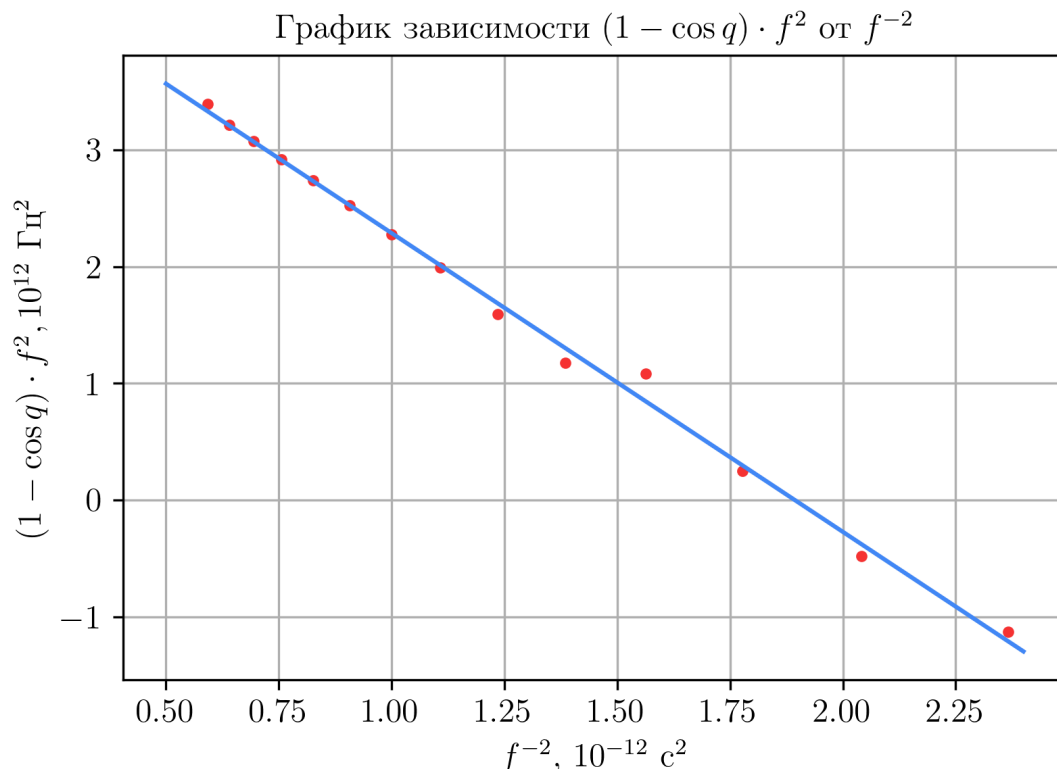
Преобразуя выражение получаем линеаризацию  $(1 - \cos q) \cdot \omega^2 L$  от  $\frac{1}{\omega^2 L}$

$$(1 - \cos q) \cdot \omega^2 L = \frac{C_1 + C_2}{C_1 C_2} - \frac{1}{2C_1 C_2} \cdot \frac{1}{\omega^2 L}.$$

We linearize the dependence and construct a graph.

Linear dependence parameters (slope coefficient and shift index, respectively)

$$k = -2.58 \cdot 10^{24} \text{ Гц}^4, \quad b = 4.86 \cdot 10^{12} \text{ Гц}^2.$$



**A6** Using the coefficients found in A5, determine the capacitances of capacitors  $C_1$  and  $C_2$  ( $C_1 < C_2$ ). **0.70**

According to the theory,

$$k = -\frac{(2\pi)^2}{2L^2} \cdot \frac{1}{C_1 C_2}, \quad b = \frac{(2\pi)^2}{L} \cdot \frac{C_1 + C_2}{C_1 C_2}.$$

From where we get (using, for example, Vieta's theorem for a quadratic equation whose roots are inverse capacitances)

$$\begin{aligned} C_1 &= 0.43 \text{ nF}, \\ C_2 &= 0.89 \text{ nF}. \end{aligned}$$

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$$\begin{aligned} C_1 &= 0.43 \text{ nF}, \\ C_2 &= 0.89 \text{ nF}. \end{aligned}$$

**B1** Express the impedances  $z_r$  and  $z_l$  for waves traveling to the right and left. Express your answer in terms of  $C_1$ ,  $C_2$ ,  $L$ ,  $q$  and  $\omega$ . **0.80**

Let's arrange the currents and potentials in the circuit similarly to part A

Then we write down the wave impedance  $z_r$  for a wave traveling to the right

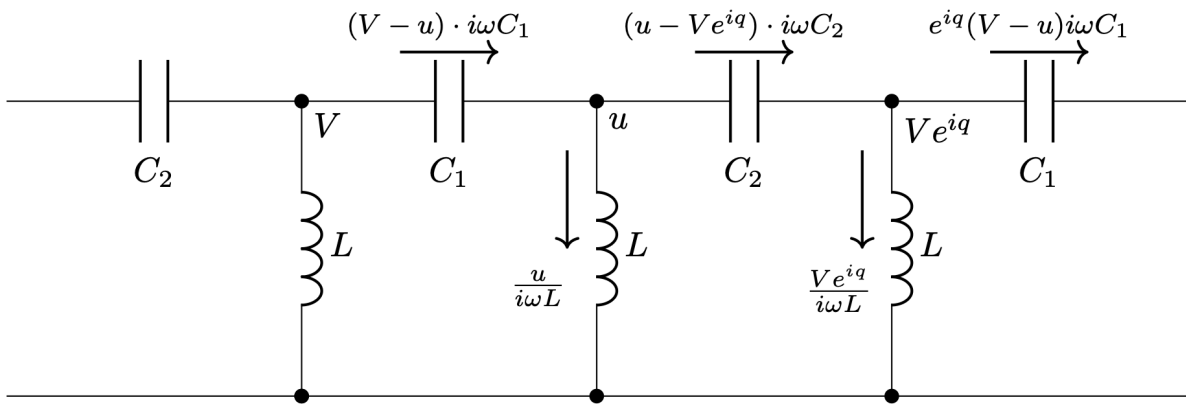
$$z_r = \frac{V}{I} = \frac{V}{V-u} \cdot \frac{1}{i\omega C_1}.$$

Подставим соотношение для  $\frac{u}{V}$ , полученное в части А

$$z_r = \frac{1}{1 - \frac{C_1 + C_2 e^{iq}}{C_1 + C_2 - \frac{1}{\omega^2 L}}} \cdot \frac{1}{i\omega C_1} = \frac{C_1 + C_2 - \frac{1}{\omega^2 L}}{C_2(1 - e^{iq}) - \frac{1}{\omega^2 L}} \cdot \frac{1}{i\omega C_1}.$$

Выражение для  $z_l$  можно получить используя замену  $q \rightarrow -q$

$$z_l = \frac{C_1 + C_2 - \frac{1}{\omega^2 L}}{C_2(1 - e^{-iq}) - \frac{1}{\omega^2 L}} \cdot \frac{1}{i\omega C_1}.$$



**B2** Write down the boundary condition connecting the complex stress amplitudes  $V_r(0, t)$  and  $V_l(0, t)$ . The answer will be an equation containing  $V_r(0, t)$ ,  $V_l(0, t)$ ,  $z_r$ ,  $z_l$ ,  $q$  and  $N$ . **0.60**

Let's write the boundary condition for the current flowing from a link in the circuit with number  $N$

$$I_r(N, t) + I_l(N, t) = 0.$$

Let's express the currents of each of the waves through the corresponding voltages on the zero link

$$\frac{V_r(0, t) e^{iqN}}{z_r} + \frac{V_l(0, t) e^{-iqN}}{z_l} = 0,$$

$$V_l(0, t) = -\frac{z_l}{z_r} V_r(0, t) e^{2iqN}.$$

**B3** Obtain an expression for the total impedance modulus of a circuit  $|z|$  consisting of  $N$  links. Express your answer using  $N$ ,  $q$  and  $|z_r|$ . **1.20**

Let's write an expression for the total impedance of the circuit by substituting the amplitude of voltage and current as the sum of the amplitudes of voltage and current of two waves

$$z = \frac{V_r(0, t) + V_l(0, t)}{I_r(0, t) + I_l(0, t)} = \frac{V_r(0, t) + V_l(0, t)}{\frac{V_r(0, t)}{z_r} + \frac{V_l(0, t)}{z_l}}.$$

Подставим полученные соотношения на амплитуды напряжений волн

$$z = \frac{1 - \frac{z_l}{z_r} e^{2iqN}}{\frac{1}{z_r} + \frac{1}{z_l} \left( -\frac{z_l}{z_r} e^{2iqN} \right)} = \frac{z_r - z_l e^{2iqN}}{1 - e^{2iqN}}.$$

Подставим соотношение на импедансы и пренебрежем  $\frac{1}{\omega^2 L}$

$$z = z_r \frac{1 - \frac{C_2 (1 - e^{iq}) - \frac{1}{\omega^2 L}}{C_2 (1 - e^{-iq}) - \frac{1}{\omega^2 L}} \cdot e^{2iqN}}{1 - e^{2iqN}} = z_r \frac{1 - \frac{1 - e^{iq}}{1 - e^{-iq}} \cdot e^{2iqN}}{1 - e^{2iqN}} = z_r \frac{1 + e^{2iq(N+\frac{1}{2})}}{1 - e^{2iqN}}.$$

Возьмем модуль от полученного выражения

$$|z| = |z_r| \frac{\cos \left( q \left( N + \frac{1}{2} \right) \right)}{\sin(qN)}.$$

**B4** Using the results obtained in Part A, write an equation for the frequencies at which the maximum impedance is achieved. Express your answer in terms of  $N$ ,  $C_1$ ,  $C_2$ ,  $L$ ,  $\omega$  and the integer parameter  $m$ . **0.60**

The maximum modulus is reached at  $qN = \pi \Delta n$ . Let us obtain an expression for the frequencies at which the impedance extrema

$$\arccos \left( \frac{1}{2C_1 C_2} \left( \frac{1}{\omega^2 L} - (C_1 + C_2) \right)^2 - (C_1^2 + C_2^2) \right) = \frac{\pi \Delta n}{N}.$$

are reached

**B5** Measure the frequencies at which the maximum impedance modulus is realized. Measurements should be carried out at frequencies  $1400 \text{ кГц} < f < 6000 \text{ кГц}$ , at which the LC circuit passes into the conduction band. **1.00**

Let's connect to the board according to the condition, we will measure the frequencies at which the minimum amplitude of the voltage across the resistor is achieved. We obtain the signal on the resistor using the math oscilloscope mode, subtracting from the signal taken from the generator the signal taken on the J2 - GND contacts.

$n \ f_{rez}, \text{ кГц}$   $q \ 1 \ 1466 \ 3.49 \ 2 \ 1590 \ 3.70 \ 3 \ 1717 \ 3.88 \ 4 \ 1920 \ 4.10 \ 5 \ 2137 \ 4.29 \ 6 \ 2489 \ 4.52 \ 7 \ 2899 \ 4.70 \ 8 \ 3579 \ 4.90 \ 9 \ 4549 \ 5.06$

Let us construct the dependence of  $q = 2\pi - \arccos\left(\frac{1}{2C_1C_2}\left(\frac{1}{\omega^2L} - (C_1 + C_2)\right)^2 - (C_1^2 + C_2^2)\right)$  on  $n$ .

The slope is equal to  $k = 0.198$ , hence  $N = 15.9 \approx 16$ .

